10.8.2 The dis_max query

In the previous section, we looked at the `multi_match` query, which searches for criteria in multiple fields. To execute this query type behind the scenes, Elasticsearch rewrites the `multi_match` query using a *disjunction max query* (`dis_max`). The `dis_max` query splits each field into a separate `match` query, as the following listing shows.

Listing 10.16 Disjunction max (`dis_max`) query in use

```
GET books/_search
{
  "_source": false,
  "query": {
    "dis_max": {
      "queries": [
        {"match": {"title": "Design Patterns"}},
        {"match": {"synopsis": "Design Patterns"}}]
    }
  }
}
```

Specifies the dis_max query type

Defines the set of queries to include in a dis_max query block

Specifies a match query

Multiple fields are split into two `match` queries under the `dis_max` query. The query returns documents with a high relevancy `_score` for the individual field.

> **NOTE** The `dis_max` query is classified as a compound query: a query that wraps up other queries. We discuss compound queries in chapter 11.

In some situations, the relevancy scores of the fields in a `multi_match` query are the same. In that case, the scores end up in a tie. To break the tie, we use a tiebreaker.

10.8.3 Tiebreakers

The relevancy score is based on a single field's score, but if the scores are tied, we can specify `tie_breaker` to solve the tie. When we use `tie_breaker`, Elasticsearch calculates the overall score slightly differently, as we see shortly; first, let's check out an example.

The following listing queries two words against two fields: `title` and `tags`. However, the code also adds a `tie_breaker` parameter.

Listing 10.17 A `multi_match` query with a tiebreaker

```
GET books/_search
{
  "query": {
    "multi_match": {
      "query": "Design Patterns",
      "type": "best_fields",
      "fields": ["title","tags"],
      "tie_breaker": 0.9
    }
  }
}
```

Specifies a multi_match query

Queries for "Design Patterns"

Sets the query's type to best_fields

Defines the set of fields to search

Sets the tiebreaker